Recap:

New notation:

Fourier Transform of function x is the function X defined as:

$$X(\omega) = \int_{t=-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} \, dt.$$

$$x \xleftrightarrow{F} X$$

I. Examples:

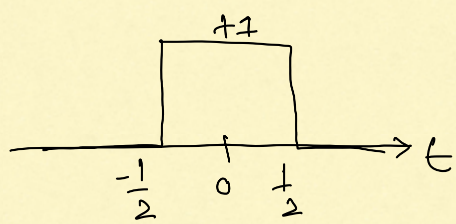
1. $x(t) = e^{-at} \cdot u(t)$, a is real and $a > 0$.

$$X(\omega) = \frac{1}{a + j\omega}$$

$$e^{-at} u(t) \xleftrightarrow{F} \frac{1}{a + j\omega}$$

2. Top hat function:

$$\Pi(t) = \begin{cases} 1 & \text{if } -\frac{1}{2} < t < \frac{1}{2} \\ 0 & \text{else} \end{cases}$$



If $x(t) = \Pi(t)$ then:

$$X(\omega) = \int_{-1/2}^{+1/2} x(t) \cdot e^{-j\omega t}\, dt$$

$$= \int_{-1/2}^{+1/2} e^{-j\omega t}\, dt.$$

(i) $\underline{\omega = 0}$: $X(\omega) = X(0) = 1$

(ii) $\underline{\omega \neq 0}$:

$$X(\omega) = \int_{-1/2}^{+1/2} e^{-j\omega t}\, dt$$

$$= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-1/2}^{+1/2}$$

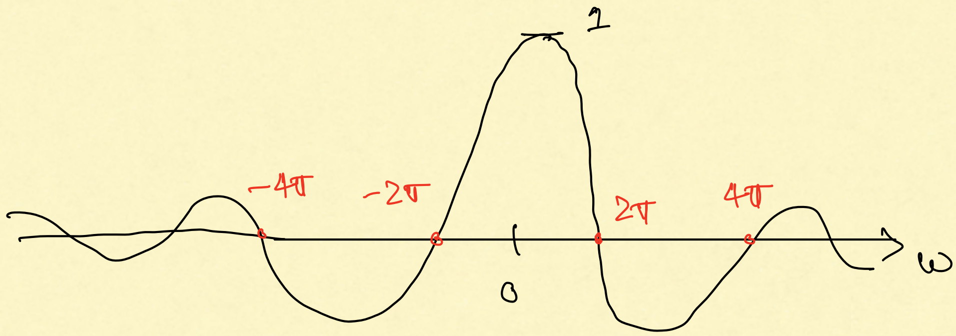
$$= \frac{1}{-j\omega} \times \left[e^{-j\omega/2} - e^{+j\omega/2} \right]$$

$$= \frac{1}{-j\omega} \times -2j \sin(\omega/2) = \frac{\sin(\omega/2)}{\omega/2}$$

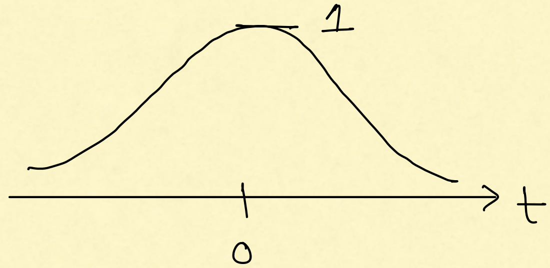
$$= \frac{\sin(\pi \cdot \omega/2\pi)}{\pi \cdot \omega/2\pi} \qquad \left\{ \operatorname{sinc}(u) = \frac{\sin \pi u}{\pi u} \right.$$

$$= \operatorname{sinc}\left(\frac{\omega}{2\pi}\right)$$

$$\Pi(t) \xleftrightarrow{F} \mathrm{sinc}\left(\frac{\omega}{2\pi}\right).$$



3. $x(t) = e^{-t^2}$



Fact 1: $$\int_{-\infty}^{+\infty} e^{-t^2}\,dt = \sqrt{\pi}.$$

Fact 2: $$\int_{-\infty}^{+\infty} e^{-(t+a)^2}\,dt = \sqrt{\pi}, \quad \text{for any } a \in \mathbb{C}.$$

$$X(\omega) = \int_{-\infty}^{+\infty} e^{-t^2} \cdot e^{-j\omega t}\,dt$$

$$= \int_{-\infty}^{+\infty} \exp\left[-(t^2 + j\omega t)\right] dt$$

$$= \int_{-\infty}^{+\infty} \exp\left[-(t + j\omega/2)^2 + (j\omega/2)^2\right] dt$$

$$= e^{(j\omega/2)^2} \cdot \int_{-\infty}^{+\infty} e^{-(t+j\omega/2)^2} dt$$

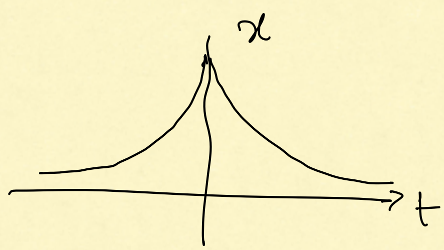
$$= e^{-\omega^2/4} \cdot \sqrt{\pi} \quad , \quad \text{using Fact 2.}$$

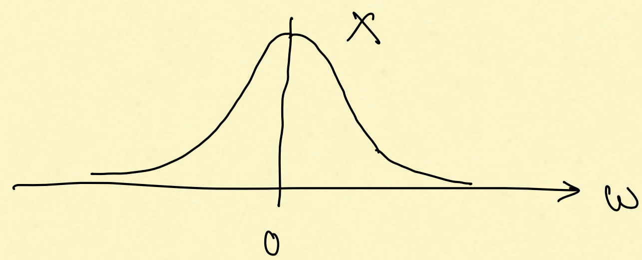
$$e^{-t^2} \xleftrightarrow{F} \sqrt{\pi} \cdot e^{-\omega^2/4}.$$

4. <u>Homework</u>

$$e^{-a|t|} \xleftrightarrow{F} \frac{2a}{a^2+\omega^2}$$

$a > 0$





II Fourier Inversion Theorem

<u>Definition</u>: A signal x is <u>absolutely integrable</u> if

$$\int_{-\infty}^{+\infty} |x(t)|\, dt \text{ is finite.}$$

Theorem: Fourier Inversion Theorem (Körner, Thm 60.1)

If both x & X are absolutely integrable and are continuous everywhere then:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) \cdot e^{j\omega t} \, d\omega, \text{ for all } t.$$

For reference $X(\omega) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} \, dt.$

Examples:

① $x(t) = e^{-a|t|}$, $a > 0$

$$X(\omega) = \frac{2a}{a^2 + \omega^2}$$

$$\frac{1}{2\pi} \int_{\omega=-\infty}^{+\infty} \frac{2a}{a^2+\omega^2} \cdot e^{j\omega t} \, d\omega = e^{-a|t|} \quad \forall t.$$

② $x(t) = e^{-t^2}$, $X(\omega) = \sqrt{\pi} \cdot e^{-\omega^2/4}$

$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} \sqrt{\pi} \cdot e^{-\omega^2/4} \cdot e^{j\omega t} \, d\omega = e^{-t^2} \quad \forall t.$$

II. Convolution Property.

$$y = h * x \implies Y(\omega) = H(\omega) \cdot X(\omega)$$

under some conditions.

Main Idea:

$$y(t) = \int_\tau h(\tau) \cdot x(t-\tau) d\tau$$

$$Y(\omega) = \int_t \left[\int_\tau h(\tau) x(t-\tau) d\tau \right] \cdot e^{-j\omega t} dt$$

under some conditions

$$\overset{*}{=} \int_\tau h(\tau) \int_t x(t-\tau) \underbrace{e^{-j\omega t}}_{e^{-j\omega(t-\tau)} \cdot e^{-j\omega\tau}} dt \cdot d\tau$$

$$= \int_\tau h(\tau) \left\{ \int_t x(t-\tau) \cdot e^{-j\omega(t-\tau)} dt \right\} \cdot e^{-j\omega\tau} d\tau$$

$$= X(\omega) \int_\tau h(\tau) \cdot e^{-j\omega\tau} d\tau$$

$$= X(\omega) \cdot H(\omega).$$

Theorem: If x and h are absolutely integrable and bounded then:

$$y = x * h \implies Y(\omega) = X(\omega) \cdot H(\omega).$$

We say that x is bounded if

$\sup_t |x(t)|$ is finite.

Another equivalent definition:

x is bounded if there exists a finite constant B such that

$$|x(t)| \leq B \text{ for all } t.$$

Absolute Integrability : Examples

- The following functions are absolutely integrable:

$\Pi(t)$, $e^{-at}u(t)$ for $a>0$, $e^{-a|t|}$ for $a>0$,

$\dfrac{2a}{a^2+\omega^2}$ (as a function of ω), $\operatorname{sinc}^2(t)$

In fact: $\displaystyle\int_{-\infty}^{+\infty} |\operatorname{sinc}^2(t)|\,dt = \int_{-\infty}^{+\infty} \operatorname{sinc}^2(t)\,dt = 1.$

- **Fact:** All Schwartz signals are absolutely integrable, continuous and bounded

- sinc(t) is not absolutely integrable.

$$\int_{-\infty}^{+\infty} |\text{sinc}(t)|\, dt = \infty .$$

- $\frac{1}{a+j\omega}$ (as a function is not absolutely integrable., for $a>0$.

$$\int_{-\infty}^{+\infty} \left|\frac{1}{a+j\omega}\right| d\omega = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{a^2+\omega^2}}\, d\omega$$

($\frac{1}{\sqrt{a^2+\omega^2}}$ decays as $\frac{1}{\omega}$ for large values of ω.)